Section 2.4

Rules for matrix operations

Homework List

Strang:

Section 2.3: 1, 3, 31

Section 2.5: 25, 30

Section 2.6: 3, 4, 7, 9, 10, 13, 14, 21

Section 2.7: 22, 34

Tucker:

Section 3.3: 8, 9, 10, 11, 14, 16

Rules of matrixalgebra.

- **Matrix addition is associative and commutative:**

$$A + (B + C) = (A + B) + C$$
$$A + B = B + A$$

- **Matrix multiplication is associative, but usually NOT commutative:**

$$A(BC) = (AB)C$$
$$AB \neq BA$$

- **Matrix multiplication and addition are distributive:**

$$A(B + C) = AB + AC$$
$$(A + B)C = AC + BC$$

- **Treat vectors as (1 x n) or (n x 1) matrices and obey all the same rules!**

Rules of matrix algebra.

- ***Scalar*** **multiplication is associative and commutative:**

$$k(A+B) = kA + kB$$
$$(k+l)A = kA + lA$$
$$kA = Ak$$

$$k(AB) = (kA)B = (Ak)B = A(kB) = A(Bk) = (AB)k$$

- **All these rules assume that the dimensions match in an appropriate way; the rules only hold when the expression makes sense.**
- **Treat vectors as (1 x n) or (n x 1) matrices and obey all the same rules!**

Rules of matrix algebra.

- **Remember the additive and multiplicative identities:**

$$\mathbf{0} + A = A + \mathbf{0} = A$$
$$IA = AI = A$$
$$\mathbf{0}A = \mathbf{0}, \qquad A\mathbf{0} = \mathbf{0}$$

- ***NOTE: The product of a zero with any matrix is also a zero matrix, but the size of the result depends on the dimensions of the zero matrix. The product of a matrix with the scalar 0 is a zero matrix of the same size.***
- **You may sometimes encounter the ones matrix (a matrix with every element equal to 1). The ones matrix doesn't really have any special algebraic rules; importantly, it is NOT the identity:**

$$\mathbf{1}A \neq A, \qquad A\mathbf{1} \neq A, \qquad \mathbf{1}A \neq A\mathbf{1}$$

Rules of matrix algebra.

- **The transpose of a matrix can be distributed into matrix addition and with into matrix multiplication.:**

$$(A+B)^T = A^T + B^T$$
$$(AB)^T = B^T A^T$$

- **Note the change of order when the transpose is distributed into a product; for non-square matrices, the sizes wouldn't match if we did not do this.**

Rules of matrix algebra.

- **The inverse of a matrix can be distributed into matrix multiplication but not matrix addition and with into matrix multiplication.:**

$$(A+B)^{-1} \neq A^{-1} + B^{-1}$$
$$(AB)^{-1} = B^{-1}A^{-1}$$

- ***NOTE: Not only does the matrix inverse not distribute into matrix addition, the inverse may not even exist for all the matrices.***
- ***On the other hand, if AB has an inverse, then so do A and B, and vice versa.***

Block matrix and block multiplication

We have to say one more thing about matrices. They can be cut into ***blocks*** (which are smaller matrices). This often happens naturally. Here is a 4 by 6 matrix broken into blocks of size 2 by 2—in this example each block is just I:

4 by 6 matrix
2 by 2 blocks give
2 by 3 block matrix

$$A = \left[\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right] = \begin{bmatrix} I & I & I \\ I & I & I \end{bmatrix}.$$

Block matrices.

- **The blocks need to be sized appropriately to make sense. An example:**

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, A \in \mathbb{R}^{p\times r}, B \in \mathbb{R}^{p\times s}, C \in \mathbb{R}^{q\times r}, D \in \mathbb{R}^{q\times s}, M \in \mathbb{R}^{(p+q)\times(r+s)}$$

- **The augmented matrix is a block matrix, $M = [A \quad \vec{b}]$; the first block is the coefficient matrix and the second is the RHS vector (both have the same number of rows.**
- **We can perform operations with block matrices much as we can with normal matrices, as long as the blocks are all sized appropriately. An example:**

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}\begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} (AE+BG) & (AF+BH) \\ (CE+DG) & (CF+DH) \end{bmatrix}$$

  - **The number of columns in A and C must match the number of rows in E and F. (Multiplication)**
  - **The number of columns in B and D must match the number of rows in G and H. (Multiplication)**
  - **The number of rows of A and C must match the number of row of B and D. (Addition)**
  - **The number of columns of E and F must match the number of columns of G and H . (Addition)**

Optional: Column and Row multiplication

This is interesting but can be confusing. So, it is optional for you to know this method.

$$\begin{bmatrix} \textbf{col 1} & \textbf{col 2} & \textbf{col 3} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} \textbf{row 1} & \cdot & \cdot & \cdot \\ \textbf{row 2} & \cdot & \cdot & \cdot \\ \textbf{row 3} & \cdot & \cdot & \cdot \end{bmatrix} = (\textbf{col 1})\,(\textbf{row 1}) + (\textbf{col 2})\,(\textbf{row 2}) + (\textbf{col 3})\,(\textbf{row 3}).$$

$$AB = \begin{bmatrix} \boldsymbol{a} & b \\ \boldsymbol{c} & d \end{bmatrix} \begin{bmatrix} \boldsymbol{E} & \boldsymbol{F} \\ G & H \end{bmatrix} = \begin{bmatrix} \boldsymbol{aE} + bG & \boldsymbol{aF} + bH \\ \boldsymbol{cE} + dG & \boldsymbol{cF} + dH \end{bmatrix}$$

Add columns of A times rows of B

$$AB = \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{c} \end{bmatrix} \begin{bmatrix} \boldsymbol{E} & \boldsymbol{F} \end{bmatrix} + \begin{bmatrix} b \\ d \end{bmatrix} \begin{bmatrix} G & H \end{bmatrix}$$

Section 2.5

The matrix inverse.

Review Identity matrices

- **Recall that an identity matrix I is always a square matrix:**

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & & 0 & 0 \\ & \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Multiplicative identities

- **Multiplication of any matrix by appropriately-sized identity matrix gives the same matrix as the result:**

$$I_m A = A I_n = A, \forall A \in \mathbb{R}^{m \times n}$$

- **When A is a square matrix, it follows that**

$$IA = AI = A, \forall A \in \mathbb{R}^{n \times n}$$

- I_n **is the multiplicative identity of all** n**-square matrices.**
- I **is analogous to the number 1:**

$$1x = x1 = x, \forall x \in \mathbb{Z}, \mathbb{Q}, \mathbb{R}$$

Multiplicative inverses

Definition: The multiplicative inverse of an n-square matrix is defined as:

$$Given\ A \in \mathbb{R}^{n\times n},\ if\ \exists B \in \mathbb{R}^{n\times n}\colon AB = BA = I,\ then\ B \equiv A^{-1}$$

Questions:

- **Does the inverse exist for non-square matrix? (No!)**
- **For any given A, is there a matrix B such that $AB = I$? (No!)**
- **If $AB = I$, must $BA = I$? (Yes!)**
- **How can we find this matrix? (Important: Gauss-Jordan)**

Answers

- **Does the inverse exist for non-square matrix? (No!)**
  - **If AB=BA, this means A and B must be square matrices of the same size. You can assume that A is mxr and B is rxn matrices. Then you can get m=n when you need to multiply BA. The r=m because AB=BA.**
- **For any given A, is there a matrix B such that $AB = I$? (No!)**
  - **You can use a matrix A with one row is all zeros. Then the inverse does not exist.**
- **If $AB = I$, must $BA = I$? (Yes!)**
  - **B=BI= B(AB) = (BA)B Then (I-BA)B =0 Thus I=BA.**

Invertible Matrix

DEFINITION The matrix A is ***invertible*** if there exists a matrix A^{-1} that "inverts" A:

Two-sided inverse $$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I. \tag{1}$$

Note 1 ***The inverse exists if and only if elimination produces n pivots***

Note 4 (Important) ***Suppose there is a nonzero vector x such that $Ax = 0$.*** *Then A cannot have an inverse.* No matrix can bring $\mathbf{0}$ back to x.

Note 5 A 2 by 2 matrix is invertible if and only if $ad - bc$ is not zero:

$$\textbf{2 by 2 Inverse:} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Note 6 A diagonal matrix has an inverse provided no diagonal entries are zero:

$$\text{If} \quad A = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix} \quad \text{then} \quad A^{-1} = \begin{bmatrix} 1/d_1 & & \\ & \ddots & \\ & & 1/d_n \end{bmatrix}.$$

Summary: Existence and uniqueness of matrix inverse

- **Not all square matrices have an inverse. If the inverse exists, it is unique (Note 2 in Strang's Book).**
- **If a square matrix DOES have an inverse, we say that it is invertible or non-singular.**
- **If a square matrix DOES NOT have an inverse, we say that it is non-invertible or singular.**

Properties of the matrix inverse

- **If an n-square matrix A is non-singular, then:**

$$AA^{-1} = A^{-1}A = I \leftrightarrow (A^{-1})^{-1} = A$$

$$(A^T)^{-1} = (A^{-1})^T$$

- **If n-square matrices, A and B, are both non-singular, then AB and BA are also non-singular and:**

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (BA)^{-1} = A^{-1}B^{-1}$$

  - **Remember, most likely $AB \neq BA$.**

- **If either A or B (or both) are singular, then AB and BA are also singular.**

Matrix inverses and linear systems

A linear system of equations can be written as:

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{n\times n}$$

- **(Note 3 in Strang's Book) If A has an inverse, then the solution can be calculated as:**

$$A\vec{x} = \vec{b} \rightarrow A^{-1}A\vec{x} = A^{-1}\vec{b} \rightarrow \vec{x} = A^{-1}\vec{b}$$

- **Note: A is the coefficient matrix, NOT the augmented matrix.**

Non-singularity and linear systems

- A square coefficient matrix means that # of equations (rows) = # of variables (columns); such systems *often* have a single unique solution:
- If a system has a unique solution for one right-hand side, it will have a unique solution for every right-hand side or even the right side is not given (Hint: Invertible). Example: Industrial production in the face of dynamic consumer demand.

If the coefficient matrix is non-singular, we can find a solution for every right-hand side ($\vec{x} = A^{-1}\vec{b}$) → the solution must be unique!

- If the coefficient matrix is singular, then the system has either no solution or many, depending on the right-hand side.

How does a change in the RHS change the solution?

- Many applications involve uncertainty in values. Example: industrial production targets are selected by *predicted* demand.
- How does the solution change for a given change in the RHS?

$$A\vec{x} = \vec{b}, \qquad A(\vec{x} + \Delta\vec{x}) = \vec{b} + \Delta\vec{b}$$

- If $\Delta\vec{b}$ is fairly small, is $\Delta\vec{x}$ also small, or is it much larger?

$$\vec{x} = A^{-1}\vec{b}, \qquad \vec{x} + \Delta\vec{x} = A^{-1}\left(\vec{b} + \Delta\vec{b}\right) = A^{-1}\vec{b} + A^{-1}\Delta\vec{b}$$

$$\vec{x} + \Delta\vec{x} = \vec{x} + A^{-1}\Delta\vec{b} \rightarrow \Delta\vec{x} = A^{-1}\Delta\vec{b}$$

- The matrix inverse transforms a difference in the right-hand side into a difference in the solution!

Important Part

Gauss-Jordan Elimination and the matrix inverse

Non-singularity and linear systems

- **For a system of n linear equations in n unknowns:**

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{n\times n}$$

- **If A is non-singular, then the system has a unique solution given by:**

$$\vec{x} = A^{-1}\vec{b}$$

- **Recall: If the coefficient matrix is singular, then the system has either no solution or many, depending on the right-hand side.**
- **Gaussian elimination is used to obtain the conclusion.**
- **What is the relationship between Gaussian Elimination and the matrix inverse?**

Gaussian (Gauss-Jordan) Elimination

For a system of m linear equations in n unknowns, we construct an $m \times (n+1)$ augmented matrix, $M = [A \quad \vec{b}]$:

- **If forward elimination gives any rows that have zeros everywhere except the last column of the augmented matrix, then there is no solution.**
- **If forward elimination gives less than n non-zero rows, then some variables must be free variables: there are many (infinite) solutions; the augmented matrix after back substitution allows us to write a form for these solutions.**
- **If forward elimination gives exactly n non-zero rows, then every variable is a pivot and there is exactly one (unique) solution; the last column of the augmented matrix *after back substitution* is the answer.**

Singular systems lead to all-zero rows (in coefficient block).

For a square system of n linear equations in n unknowns, working with $M = [A \quad \vec{b}]$:

- **No solution (singular):** **Forward elimination gives at least one row with zeros everywhere in the coefficient block (non-zero only in last column).**
- **Many solutions (singular):** **Forward elimination gives at least one row with zeros everywhere in the coefficient block (less than n non-zero rows).**
- **One unique solution (non-singular):** **Forward elimination does not give any zero rows in the coefficient block (exactly n non-zero rows).**
- **The existence of a zero row in the coefficient block of a row echelon form is both necessary and sufficient for classifying the system as singular.**

What is a singular matrix?

- **"Most" square matrices are non-singular, why?**

Row picture:

- **A square system is *n* hyperplanes in an *n*-D space: 2 lines in 2-D, 3 planes in 3-D, etc. The coefficients represent their orientation (example for rotation matrix); the right-hand sides define the offset relative to the origin. There is a single unique solution for all offsets unless there is some special relationship between the orientation of the hyperplanes.**
- **If two rows are identical or related by a scalar multiple, the matrix must be singular. If one row is a linear combination of other rows, the matrix must be singular. If any row is zero, the matrix must be singular.**

What is a singular matrix?

- **"Most" square matrices are non-singular, why?**

Column picture:

- **Each column is a vector in an n-D space. We have n vectors and would like to take their linear combinations: 2 vectors in 2-D, 3 vectors in 3-D, etc. Unless there is a special relationship between the columns, the set of all linear combinations is the entire n-D space; we can find a unique solution for any vector we place on the right-hand side.**
- **If two columns are identical or related by a scalar multiple, the matrix must be singular. If one column is a linear combination of the others, the matrix must be singular. If any column is zero, the matrix must be singular.**

Non-singular systems lead to the identity matrix.

For a non-singular square system of n linear equations in n unknowns, working with $M = [A \quad \vec{b}]$:

- **There are exactly n non-zero rows in any echelon form.**
- **Every variable is a pivot.**
- **There are no free variables.**
- **Back substitution leads to a reduced row echelon form with ones on the diagonal and zeros everywhere else, except for the last column.**
- **The coefficient block of the reduced row echelon form of non-singular system is the $n \times n$ identity matrix!**

Elementary matrices.

- **Recall three elementary row operations in Gaussian Elimination:**
  - **Addition of a scalar multiple:** $R_i + kR_j \rightarrow R_i$
  - **Scalar multiplication:** $kR_i \rightarrow R_i$
  - **Row exchange:** $R_i \leftrightarrow R_j$
- **The elementary matrices are generated by applying the same operation to the identity matrix. Each row operation corresponds to an elementary matrix.**
  - **Each of these operations can be achieved through matrixmultiplication by an elementary matrix, i.e., if we want to exchange two rows of a Matrix A, then we can multiply an elementary matrix E that exchanges the same two rows on the identity matrix from the left side (EA).**

Example elementary matrices.

- **Row exchange:** $R_i \leftrightarrow R_j$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = A'$$

Example elementary matrices.

- **Addition of a scalar multiple:** $R_i + kR_j \to R_i$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \to R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \to R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

Example elementary matrices.

- **Scalar multiplication:** $kR_i \to R_i$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{{}^1/_2 R_2 \to R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & {}^1/_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} \xrightarrow{{}^1/_2 R_2 \to R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & {}^1/_2 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & {}^1/_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & {}^1/_2 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

Gaussian Elimination with matrices.

- **We start with a sequence of r row operations, forward elimination, that leads to a row echelon (upper triangular) form:**

$$M \xrightarrow{R_{f1}} M_{f1} \xrightarrow{R_{f2}} M_{f2} \xrightarrow{R_{f3}} \cdots \xrightarrow{R_{fr-1}} M_{fr-1} \xrightarrow{R_{fr}} M_{fr} = U$$

- **Each operation can be achieved by multiplication by an elementary matrix:**

$$M \rightarrow E_{f1}M \rightarrow E_{f2}(E_{f1}M) \rightarrow \cdots \rightarrow E_{fr}\left(E_{fr-1}\left(\cdots E_{f2}(E_{f1}M)\right)\right) = U$$

- **Forward elimination becomes a product of elementary matrices!**

$$E_{fr}\left(E_{fr-1}\left(\cdots E_{f2}(E_{f1}M)\right)\right) = \left(\prod_{i=0}^{r-1} E_{fr-i}\right) M = FM = U, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i}\right)$$

Gaussian Elimination with matrices.

- **We continue with a sequence of s row operations, back substitution, that leads to the reduced row echelon form:**

$$U \xrightarrow{R_{b1}} U_{b1} \xrightarrow{R_{b2}} U_{b2} \xrightarrow{R_{b3}} \cdots \xrightarrow{R_{bs-1}} U_{bs-1} \xrightarrow{R_{bs}} U_{bs} = R$$

- **Again, each operation can be achieved by multiplication by an elementary matrix:**

$$U \rightarrow E_{b1}U \rightarrow E_{b2}(E_{b1}U) \rightarrow \cdots \rightarrow E_{bs}\left(E_{bs-1}\left(\cdots E_{b2}(E_{b1}U)\right)\right) = R$$

- **Back substitution also becomes a product of elementary matrices!**

$$E_{bs}\left(E_{bs-1}\left(\cdots E_{b2}(E_{b1}U)\right)\right) = \left(\prod_{i=0}^{s-1} E_{bs-i}\right)U = BU = R, \qquad B = \left(\prod_{i=0}^{s-1} E_{bs-i}\right)$$

Gaussian elimination is just a matrix product!

$$M \to \cdots \to U \to \cdots \to R$$

$$U = FM, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right)$$

$$R = BU, \qquad B = \left(\prod_{i=0}^{s-1} E_{bs-i} \right)$$

$$R = B(FM) = (BF)M$$

Gaussian Elimination with matrices.

- **If the inverse of Matrix A exists, then M is an augmented matrix of a non-singular square system.**
- **It follows that the reduced row echelon form R has the identity matrix in the coefficient block and the solution in the last column.**

$$M = [A \quad \vec{b}], \qquad R = [I \quad \vec{x}]$$

$$R = (BF)M$$

$$[I \quad \vec{x}] = (BF)[A \quad \vec{b}] = \left[(BF)A \quad (BF)\vec{b}\right]$$

$$I = (BF)A, \qquad \vec{x} = (BF)\vec{b}$$

$$A^{-1}A = I, \qquad \vec{x} = A^{-1}\vec{b}$$

- (BF) **is the inverse of** A**!**

Matrix inverse by Gauss-Jordan Elimination

Use forward elimination and back substitution to obtain the inverse matrix A^{-1}.

Consider: $M = [A \quad I]$ Important construction!!!! Key!!!!

After full elimination: $R = (BF)M = [(BF)A \quad (BF)I]$

But: $(BF)A = I, \quad (BF)I = BF = A^{-1}$

$$[A \quad I] \xrightarrow{r\ steps\ of\ Forward\ Elimination} [U \quad X] \xrightarrow{s\ steps\ of\ back\ substitution} [I \quad A^{-1}]$$

Key: From the discussions above, we need to go through forward elimination and back substitution to get identify matrix I from A. It is equivalent to multiply BF (multiplications of all elementary matrices for the corresponding operations) on the left side of matrix A. BF is also the inverse of A.

Matrix inverse by Gauss-Jordan Elimination.

$$[A \quad I] \xrightarrow{r\ steps\ of\ Forward\ Elimination} [U \quad X] \xrightarrow{s\ steps\ of\ back\ substitution} [\text{0 row} \quad \begin{matrix}\text{Inverse}\\ \text{DNE}\end{matrix}]$$

- **If a zero row ever appears in A, then A is singular and has no inverse.**
- ***Note 1*** **: A is the coefficient matrix, NOT the augmented matrix.**
- ***Note 2*** **: A must be a square matrix and I must be the square matrix of the same size.**

Computing Examples

Examples computing matrix inverses.

Find the inverse.

- **Find the inverse of the following non-singular matrices:**

$$(a)\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \quad (b)\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \quad (c)\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the inverse.

$$(a)A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - {}^{3}/_{2} R_1 \rightarrow R_2} \begin{bmatrix} 2 & 1 & 1 & 0 \\ 0 & {}^{1}/_{2} & -{}^{3}/_{2} & 1 \end{bmatrix}$$

$$\xrightarrow{2R_2 \rightarrow R_2} \begin{bmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{bmatrix} \xrightarrow{R_1 - R_2 \rightarrow R_1} \begin{bmatrix} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{bmatrix}$$

$$\xrightarrow{{}^{1}/_{2} R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse.

$$(b)\ A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3 - R_1 \to R_3]{R_2 - R_1 \to R_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \to R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \xrightarrow[R_2 - R_3 \to R_2]{R_1 - R_3 \to R_1} \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 - R_2 \to R_1} \begin{bmatrix} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse.

$$(c)\ A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 3 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3-2R_4\to R_3]{\substack{R_1-4R_4\to R_1 \\ R_2-3R_4\to R_2}} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 & 0 & -4 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow[R_2-2R_3\to R_2]{R_1-3R_3\to R_1} \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1-2R_2\to R_1} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse, if possible.

- **Find the inverse of the following matrices or explain why the inverse can not exist.**

$$(a)\begin{bmatrix}1 & 2 & 3\\3 & 2 & 1\\2 & 0 & -2\end{bmatrix}\ (b)\begin{bmatrix}1 & 2 & 1\\1 & 3 & 2\\-1 & -1 & 0\end{bmatrix}\ (c)\begin{bmatrix}0 & 1 & 2\\1 & 2 & 3\\0 & 2 & 2\end{bmatrix}\ (d)\begin{bmatrix}1 & -2 & 1 & -2 & 1\\1 & 3 & 5 & 7 & 9\\8 & 6 & 4 & 2 & 0\\3 & 3 & 3 & 3 & 3\end{bmatrix}$$

Find the inverse, if possible.

$$(a)\ A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 0 & -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3 - 2R_1 \to R_3]{R_2 - 3R_1 \to R_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -4 & -8 & -3 & 1 & 0 \\ 0 & -4 & -8 & -2 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \to R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -4 & -8 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix}$$

- **There is no inverse.**
- **We have encountered a zero row in the coefficient block.**

Find the inverse, if possible.

$$(b)\ A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & -1 & 0 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3+R_1\to R_3]{R_2-R_1\to R_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3-R_2\to R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -1 & 1 \end{bmatrix}$$

- **There is no inverse.**
- **We have encountered a zero row in the coefficient block.**

Find the inverse, if possible.

$$(c)\ A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} -1 & 1 & -1/2 \\ -1 & 0 & 1 \\ 1 & 0 & -1/2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - 2R_2 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix} \xrightarrow{-1/2 R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix}$$

$$\xrightarrow[R_2 - 2R_3 \rightarrow R_2]{R_1 - 3R_3 \rightarrow R_1} \begin{bmatrix} 1 & 2 & 0 & -3 & 1 & 3/2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix} \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & -1 & 1 & -1/2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse, if possible.

$$(d)\begin{bmatrix} 1 & -2 & 1 & -2 & 1 \\ 1 & 3 & 5 & 7 & 9 \\ 8 & 6 & 4 & 2 & 0 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$$

- **There is no inverse.**
- **This is a 4×5 matrix; the inverse can only exist for square matrices.**

Identify the source of singularity.

- **Without using Gauss-Jordan elimination, explain why the following matrices are singular:**

$$(a)\begin{bmatrix}1 & 0 & 1\\ 1 & 1 & 1\\ 2 & 0 & 2\end{bmatrix}\ (b)\begin{bmatrix}1 & 0 & -1\\ 2 & 0 & 3\\ 1 & 0 & 2\end{bmatrix}\ (c)\begin{bmatrix}1 & -2 & 1 & -2 & 1\\ -2 & 1 & -2 & 1 & -2\\ 3 & 2 & 1 & 2 & 3\\ 4 & 4 & 4 & 4 & 4\\ 3 & -4 & 5 & -4 & 3\end{bmatrix}$$

Identify the source of singularity.

$$(a)\begin{bmatrix}1 & 0 & 1\\ 1 & 1 & 1\\ 2 & 0 & 2\end{bmatrix}\ (b)\begin{bmatrix}1 & 0 & -1\\ 2 & 0 & 3\\ 1 & 0 & 2\end{bmatrix}\ (c)\begin{bmatrix}1 & -2 & 1 & -2 & 1\\ -2 & 1 & -2 & 1 & -2\\ 3 & 2 & 1 & 2 & 3\\ 4 & 4 & 4 & 4 & 4\\ 3 & -4 & 5 & -4 & 3\end{bmatrix}$$

(a) The first and third columns are identical, and the third row is simply twice the first.

(b) The second column is all zeros.

(c) The first and fifth column are identical, as are the second and third columns.

Applications

Examples solving real-world problems with matrix inverses.

Industrial production.

- Several students are starting up a company to make submersible drones for underwater photography and are currently planning out how to address uncertainty in their supply chain.
- They have three models that they will produce, all of which use the same external body but differ in internal design.
  - UUV-Lite has a single propulsion unit and a single forward-facing camera.
  - UUV-Pro has a two propulsion units and a two cameras, for bi-directional movement and viewing.
  - UUV-Pro-X also has two propulsion units but has three cameras that allow for omni-directional viewing.
- If their suppliers can provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How should the production targets be changed if body production becomes more efficient, so 600 can be supplied? What if the camera supply is 1100 (with 600 bodies and 1000 propulsion units)?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

Industrial production.

- Several students are starting up a company to make submersible drones for underwater photography and are currently planning out how to address uncertainty in their supply chain.
- They have three models that they will produce, all of which use the same external body but differ in internal design.
  - UUV-Lite has a single propulsion unit and a single forward-facing camera.
  - UUV-Pro has a two propulsion units and a two cameras, for bi-directional movement and viewing.
  - UUV-Pro-X also has two propulsion units but has three cameras that allow for omni-directional viewing.
- We have multiple supply totals to consider, so we should formulate this problem using symbolic constants rather than numbers!

- Let x_i be the total number of each model to be produced: *i=1* for UUV-Lite; *i=2* for UUV-Pro; and *i=3* for UUV-Pro-X.
- Let b_j be the available supply for each component: *j=1* for bodies; *j=2* for propulsion units; and *i=3* for cameras.
- We get one equation for each component:

$$\begin{array}{rllll} Bodies: & x_1 & +x_2 & +x_3 & = b_1 \\ Propulsion: & x_1 & +2x_2 & +2x_3 & = b_2 \\ Cameras: & x_1 & +2x_2 & +3x_3 & = b_3 \end{array}$$

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = A^{-1}\vec{b}$$

- **Compute the inverse.**

Industrial production.

$$[A \quad I] = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3 - R_1 \to R_3]{R_2 - R_1 \to R_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \to R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \xrightarrow[R_2 - R_3 \to R_2]{R_1 - R_3 \to R_1} \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 - R_2 \to R_1} \begin{bmatrix} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} = [I \quad A^{-1}]$$

Industrial production.

- If their suppliers are able to provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, with the same number of propulsion units and cameras.
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, 1000 propulsion units and 1100 cameras?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = A^{-1}\vec{b}, A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 500 \\ 1000 \\ 1200 \end{bmatrix} = \begin{bmatrix} 0 \\ 300 \\ 200 \end{bmatrix}$$

- Producing 300 units of the ‘Pro’ and 200 of the ‘Pro-X’ would use no waste.

Industrial production.

- If their suppliers can provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, with the same number of propulsion units and cameras.
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, 1000 propulsion units and 1100 cameras?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

$$\vec{x} = A^{-1}\vec{b}, A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 600 \\ 1000 \\ 1200 \end{bmatrix} = \begin{bmatrix} 200 \\ -400 \\ 200 \end{bmatrix}$$

- Not a feasible solution!
- There is no production schedule that will no lead to waste with these supplies.

Industrial production.

- If their suppliers can provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, with the same number of propulsion units and cameras.
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, 1000 propulsion units and 1100 cameras?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

$$\vec{x} = A^{-1}\vec{b}, A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 600 \\ 1000 \\ 1100 \end{bmatrix} = \begin{bmatrix} 200 \\ 300 \\ 100 \end{bmatrix}$$

- **Producing 200 units of the 'Lite', 300 units of the 'Pro' and 100 of the 'Pro-X' would use no waste.**

Industrial production.

- If their suppliers can provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, with the same number of propulsion units and cameras.
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, 1000 propulsion units and 1100 cameras?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

$$A(\vec{x} + \Delta\vec{x}) = (\vec{b} + \Delta\vec{b})$$
$$(\vec{x} + \Delta\vec{x}) = A^{-1}(\vec{b} + \Delta\vec{b})$$
$$\Delta\vec{x} = A^{-1}\Delta\vec{b}$$

$$A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\Delta\vec{b}_1 = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}, \Delta\vec{b}_2 = \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix}, \Delta\vec{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}$$

Industrial production.

- If their suppliers can provide 500 bodies, 1000 propulsion units and 1200 cameras, how many of each model will they be able to produce with no waste?
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, with the same number of propulsion units and cameras.
- How many of each model will they be able to produce with no waste, if they are able to obtain 600 bodies, 1000 propulsion units and 1100 cameras?
- How does a 10-unit difference in supply of each component (independently) affect ideal (no-waste) production targets?

$$\Delta\vec{x}_1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} +20 \\ -10 \\ 0 \end{bmatrix}$$

$$\Delta\vec{x}_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix} = \begin{bmatrix} -10 \\ +20 \\ -10 \end{bmatrix}$$

$$\Delta\vec{x}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ +10 \end{bmatrix}$$

- A 10-unit increase in body supply leads to a 20-unit increase of the ‘Lite’ production target, and a 10-unit decrease in the ‘Pro’ target.
- A 10-unit increase in propulsion unit supply leads to a 10-unit decrease of both the ‘Lite’ and ‘Pro-X’ production targets and a 20-unit increase in the ‘Pro’ target.
- A 10-unit increase in camera supply leads to a 10-unit decrease of the ‘Pro’ production target, and a 10-unit increase in the ‘Pro-X’ target.

Economic equilibrium.

- **A coarse model of a national economy breaks economic activity into three sectors: primary (agriculture and natural resource extraction); secondary (manufacturing); and tertiary (services).**
- **Consider a "balanced" economy, where to the production of $1M of output from any sector requires $0.3M of input from each of the three sectors.**
- **If there is an additional (e.g. consumer) demand for $100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.**
- **If there is $150M in demand for primary and secondary sector output, but $200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?**
- **If there the consumer demand values have an uncertainty of +/- $10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?**

Economic equilibrium.

- A coarse model of a national economy breaks economic activity into three sectors: primary (agriculture and natural resource extraction); secondary (manufacturing); and tertiary (services).
- Consider a "balanced" economy, where to the production of $1M of output from any sector requires $0.3M of input from each of the three sectors.
- **We have multiple different consumer demands to consider, so we should formulate this problem using symbolic constants rather than numbers!**

- Let x_i and d_i be the total supply and demand, respectively (in millions of dollars), by economic sector *i*.
- Let b_i be the consumer demand (in millions of dollars) for the product of economic sector *i*.
- Let the sectors be defined as: agriculture & natural resources *i=1*; manufacturing *i=2*; and services *i=3*.

$$\begin{array}{lllll} d_1 = & 0.3x_1 & +0.3x_2 & +0.3x_3 & +b_1 \\ d_2 = & 0.3x_1 & +0.3x_2 & +0.3x_3 & +b_2 \\ d_3 = & 0.3x_1 & +0.3x_2 & +0.3x_3 & +b_3 \end{array}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

Economic equilibrium.

- Let x_i and d_i be the total supply and demand, respectively (in millions of dollars), by economic sector *i*.
- Let b_i be the consumer demand (in millions of dollars) for the product of economic sector *i*.
- Let the sectors be defined as: agriculture & natural resources *i=1*; manufacturing *i=2*; and services *i=3*.

$$\begin{aligned} d_1 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_1 \\ d_2 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_2 \\ d_3 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_3 \end{aligned}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

$$\vec{x} = A\vec{x} + \vec{b}, A = \begin{bmatrix} 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$M = (I - A) = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}$$

$$M\vec{x} = \vec{b}$$
$$\vec{x} = M^{-1}\vec{b}$$

- **Compute the inverse.**

Economic equilibrium.

$$[M \quad I] = \begin{bmatrix} +0.7 & -0.3 & -0.3 & 1 & 0 & 0 \\ -0.3 & +0.7 & -0.3 & 0 & 1 & 0 \\ -0.3 & -0.3 & +0.7 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[10R_3 \to R_3]{\substack{10R_1 \to R_1 \\ 10R_2 \to R_2}} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ -3 & +7 & -3 & 0 & 10 & 0 \\ -3 & -3 & +7 & 0 & 0 & 10 \end{bmatrix}$$

$$\xrightarrow[R_3 + {}^3/_7 R_1 \to R_3]{R_2 + {}^3/_7 R_1 \to R_2} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & \frac{40}{7} & \frac{-30}{7} & \frac{30}{7} & 10 & 0 \\ 0 & \frac{-30}{7} & \frac{40}{7} & \frac{30}{7} & 0 & 10 \end{bmatrix} \xrightarrow[{}^7/_{10} R_3 \to R_3]{{}^7/_{10} R_2 \to R_2} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & -3 & 4 & 3 & 0 & 7 \end{bmatrix}$$

$$\xrightarrow{R_3 + {}^3/_4 R_2 \to R_3} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & \frac{7}{4} & \frac{21}{4} & \frac{21}{4} & 7 \end{bmatrix} = [U \quad X]$$

Economic equilibrium.

$$[U \quad X] = \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & \frac{7}{4} & \frac{21}{4} & \frac{21}{4} & 7 \end{bmatrix} \xrightarrow{4/7 R_3 \to R_3} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix}$$

$$\xrightarrow[R_2+3R_3 \to R_2]{R_1+3R_3 \to R_1} \begin{bmatrix} +7 & -3 & 0 & 19 & 9 & 12 \\ 0 & 4 & 0 & 12 & 16 & 12 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} \xrightarrow{1/4 R_2 \to R_2} \begin{bmatrix} +7 & -3 & 0 & 19 & 9 & 12 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix}$$

$$\xrightarrow{R_1+3R_2 \to R_1} \begin{bmatrix} +7 & 0 & 0 & 28 & 21 & 21 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} \xrightarrow{1/7 R_1 \to R_1} \begin{bmatrix} 1 & 0 & 0 & 4 & 3 & 3 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} = [I \quad M^{-1}]$$

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for $100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is $150M in demand for primary and secondary sector output, but $200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- $10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M\vec{x} = \vec{b}, M = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = M^{-1}\vec{b}, M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix} = \begin{bmatrix} 1000 \\ 1000 \\ 1000 \end{bmatrix}$$

- Each sector must produce $1B of output.

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for $100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is $150M in demand for primary and secondary sector output, but $200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- $10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M\vec{x} = \vec{b}, M = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = M^{-1}\vec{b}, M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 150 \\ 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 1650 \\ 1650 \\ 1700 \end{bmatrix}$$

- The primary and secondary sectors must produce $1.65B of output, while the tertiary must produce $1.7B.

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for $100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is $150M in demand for primary and secondary sector output, but $200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- $10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$M(\vec{x} + \Delta\vec{x}) = (\vec{b} + \Delta\vec{b})$$
$$(\vec{x} + \Delta\vec{x}) = M^{-1}(\vec{b} + \Delta\vec{b})$$
$$\Delta\vec{x} = M^{-1}\Delta\vec{b}$$

$$\Delta\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix}$$

- Uncertainties of +/- $10M in consumer demand leads to uncertainties of +/- $100M in total equilibrium supply and demand.